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Focus Article

Heat transfer in phase change materials with multiple-fin insertion[(a)]

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Abstract – We leverage 3D numerical simulations to study phase change materials (PCMs) cells under the effect of buoyancy forces. The solid PCM is heated from a source boundary, triggering melting. The source features multiple solid fins that protrude into the PCM cell; the impact of the fins and their number is investigated by designing and testing equivalent (in terms of heating power) finless and single-fin simulations. For each configuration, the performance is quantified via the total molten substance in time. The designs were also tested for different values of the non-dimensional numbers encoding relevant properties. We confirm that fins increase the melting performance and find that single fin configurations are sub-optimal since a layout with multiple fins takes advantage of interstitial spaces, melting the substance more efficiently. The results also indicate that fins should be properly spaced, as closeness can result in overlapping, thus interfering, molten areas.

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Introduction. – Due to their large latent heat, phase change materials (PCMs) prove exceptionally useful in storing heat in many fields, including cold energy storage [1,2], building insulation [3], thermal management in electronics [4] and hydride thermal management for hydrogen adsorption/release [5,6], to cite but few examples. The dynamics of phase change in a PCM system involve a complex coupling between buoyancy forces, melting and thermal convection [7–10]: a solid substance melts due to the heating from a source boundary; the liquid, then, develops buoyancy-driven thermal convection, eventually leading to the complete liquefaction of the solid phase (see fig. 1).

Since PCM heat transfer properties are usually suboptimal, many research efforts have been devoted to relieve this flaw. Among the many design solutions in the literature [11,12], the insertion of fins delivers a simple, both conceptually and manufacturing-wise, yet effective way of transferring heat to a PCM. Paramount to fin effectiveness is their strategic placement, to promote diffusion in the early stages of melting, as well as convection in the late dynamics [4,13–15]. Recent examples in the literature include the optimization of branching fin structures to boost the performance of a battery pack thermal management system [4], a parametric study focusing on fin orientation and porosity [13] and an investigation on the effect of fin shape [14]. In [16] the trade-off between the increase in system mass and volume associated with honeycomb fins is discussed, as well as the resulting improvement in hydrogen absorption/desorption rates; in [17], a metal hydride-PCM reactor is optimized by introducing a "fin factor" index to balance heat transfer surface enhancement against geometric penalties, with the aim of improving thermally limited charging and discharging

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times. Further applications and optimizations can be found in [18–20].

Even without buoyancy, melting is a complex simulation task, due to the presence of the moving boundary between the solid and liquid phases; including gravitational effects adds the non-linearity of convection to the evolution of the interphase boundary. Thus, an accurate characterization of the temporal evolution of the melting front requires detailed local information. While such insight can be (partially) obtained through carefully designed experiments [10,21], numerical simulations grant deeper insight, allowing higher spatial and temporal resolution for the evaluation of PCM performance [9,22–24]. On the top of this quite intricate scenario, the addition of fins modifies the boundary shape, adding another layer of complexity. In this letter, we conduct numerical simulations to provide a quantitative insight on the impact of different geometrical configurations of fins in the enclosure, extending to three dimensions previous analyses conducted with the same numerical approach [15]. The extension to three dimensions is closer to physical reality, *e.g.*, simulations in three dimensions may be prone to dynamical instabilities of the fully developed flow that would be absent in the two-dimensional case. Practically, the use of three dimensions allows to have more "degrees of freedom" in positioning and sizing of the heating elements; in particular, we leverage this to address key issues on the importance of the number of adopted fins, as well as their spatial layout, focusing on the effectiveness of heating surfaces *vs.* interstitial spaces on the plane $\perp \vec{g}$. Our systematic analysis hinges on the study of layouts with the same heating areas but with different numbers of heating elements and different geometrical designs. The parametric tests are based on non-dimensional numbers characterizing the flow, providing a useful, technical operational tool: in order to dissect the effective role of fluid properties and heating element layouts, we test the same configurations with higher or lower relative convection strength, regulated by different values of the Rayleigh number Ra. Later, we provide an analysis in the conductive regime for different values of the Stefan number (*i.e.*, the ratio of sensible heat to latent heat) St.

Hydrodynamic description of thermal convection with melting. – The PCM cell is modeled as a cubic enclosure of side length L initially filled with a solid material at temperature T_{C}. The solid material is heated from a hot source on the left wall at temperature T_{H}; the side wall also presents different heating elements (multiple fins or a plate) protruding into the bulk of the enclosure (see fig. 1). The heat exchange and melting trigger the convective dynamics [21,23,25]: this complex scenario is studied at the hydrodynamical level via the continuum fields of density $\rho = \rho(\vec{x}, t)$, velocity $\vec{u} = \vec{u}(\vec{x}, t)$ and temperature $T = T(\vec{x}, t)$; melting is locally monitored via the liquid fraction $\phi = \phi(\vec{x}, t)$ that assumes values between 0.0 (solid) and 1.0 (liquid). These fields obey [8,21,26,27] the Navier-Stokes equations with buoyancy forces,

$$\partial_t \rho + \vec{\nabla} \cdot (\rho \vec{u}) = 0, \tag{1}$$

$$(\partial_t + \vec{u} \cdot \vec{\nabla})\vec{u} = \nu \nabla^2 \vec{u} - \frac{\vec{\nabla} p}{\rho} - \alpha (T - T_0)\vec{g}, \tag{2}$$

where $p = p(\vec{x}, t)$ represents the fluid pressure and ν the fluid kinematic viscosity, plus an advection-diffusion equation for the temperature featuring a melting term

$$\partial_t T + \vec{\nabla} \cdot (T\vec{u}) = \kappa \nabla^2 T - \frac{L_f}{C_p} \partial_t \phi \tag{3}$$

with κ the thermal diffusivity, L_f the latent heat of fusion and C_p the specific heat. The buoyancy term in the r.h.s. of eq. (2) is written in the Boussinesq approximation [21,26,27], being proportional to the thermal expansion coefficient α, the temperature difference with respect to a reference temperature T_0, and the gravity acceleration $\vec{g} = -g\hat{z}$ (see fig. 1). The melting term in the r.h.s. of eq. (3) models heat transfer due to the melting of the substance. The liquid fraction ϕ depends on the local enthalpy $h = h(\vec{x}, t)$ via a smoothed step function [21,24]:

$$\phi(\vec{x}, t) = \begin{cases} 0, & h < h_s, \\ \dfrac{h(\vec{x}, t) - h_s}{h_l - h_s}, & h_s \leq h \leq h_l, \\ 1, & h > h_l, \end{cases} \tag{4}$$

where $h_s = C_p T_{\mathrm{C}}$ is the solid phase enthalpy and $h_l = h_s + L_f$ the liquid phase enthalpy. The local enthalpy, in turn, depends on both the temperature and the liquid fraction, $h(\vec{x}, t) = C_p T(\vec{x}, t) + L_f \phi(\vec{x}, t)$. Regarding the boundaries, from the thermal point of view the fins and the wall from which the fins protrude are sources with a prescribed fixed temperature T_{H}; all the other boundaries of the cubic enclosure are set as insulating (*i.e.*, *adiabatic*) walls. All solid boundaries are treated as no-slip boundaries for the velocity field.

The numerical simulations of (1)–(3) are performed by means of the lattice Boltzmann model (LBM), a numerical tool that has proven to be reliable and accurate for the study of complex flow dynamics [28–32]. In a nutshell, the lattice Boltzmann equation in the BGK formulation with Guo forcing [29,33] reads

$$f_i(\vec{x} + \vec{c}_i, t + 1) - f_i(\vec{x}, t) = \\ - \frac{1}{\tau}(f_i - f_i^{(\mathrm{eq})})\,(\vec{x}, t) + \left(1 - \frac{1}{2\tau}\right) S_i(\vec{x}, t), \tag{5}$$

where $f_i(\vec{x}, t)$ is a population, *i.e.*, the probability of finding a mesoscale particle at $(\vec{x}, \vec{c}_i, t)$ in the phase-space. The velocities belong to a discrete set, the D3Q19 [28,29] in this case. In eq. 5 f_i is relaxed with characteristic time τ to a Maxwell-Boltzmann equilibrium $f_i^{(\mathrm{eq})}$ [28,29] and forces are encoded by the source term $S_i(\vec{x}, t)$ [33]. The macroscopical quantities are recovered by $\rho = \sum f_i$, $\rho\vec{u} = \sum f_i \vec{c}_i$, $\nu = c_s^2(\tau - 1/2)$, with $c_s^2 = 1/3$ being the speed

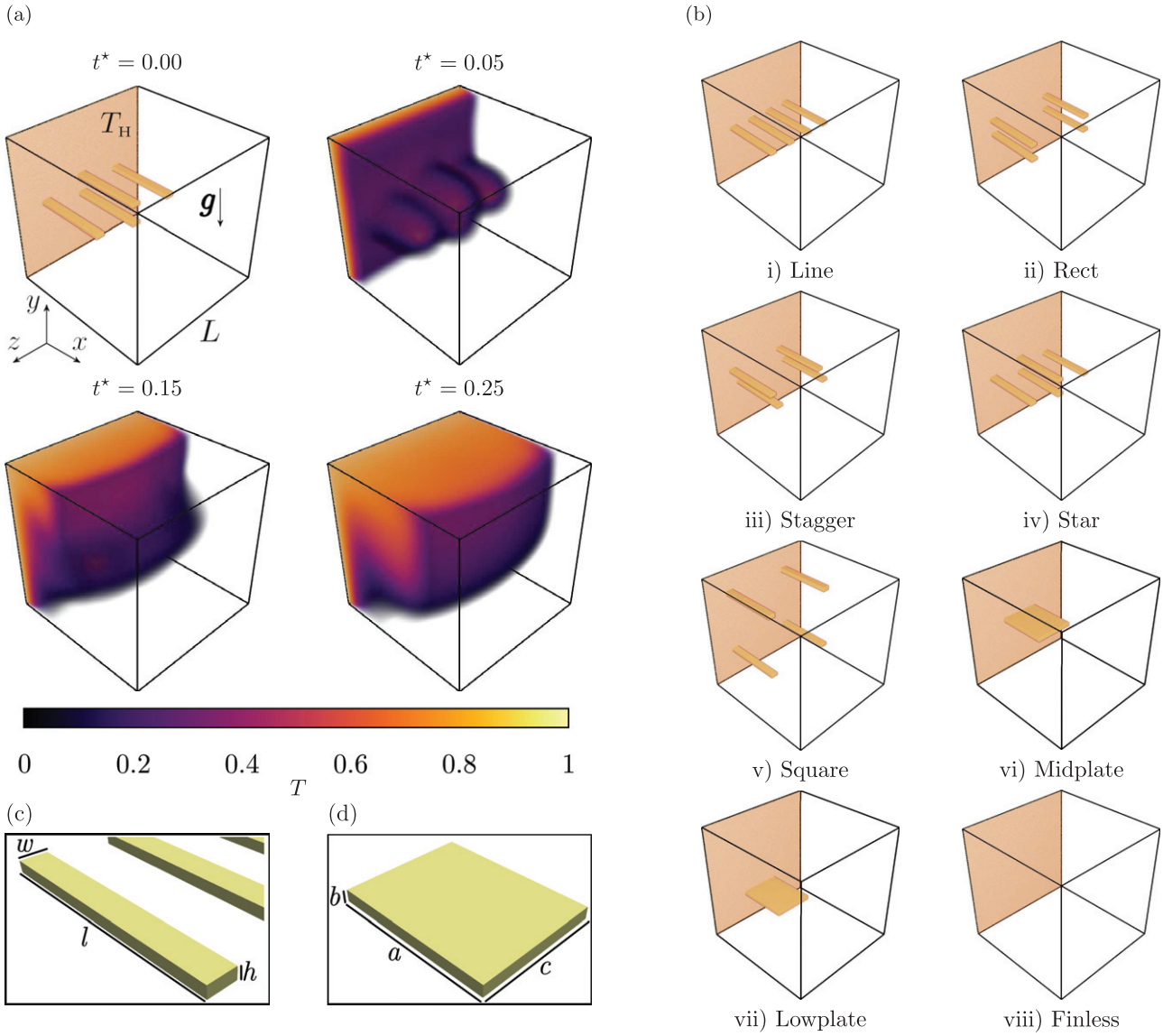


Fig. 1: Panel (a): typical snapshots of temperature field in the melting dynamics of a PCM cell consisting of a cubic enclosure with side L under the effect of gravity $\vec{g} = -g\hat{z}$. The solid substance is rendered as transparent. A side boundary of the enclosure consists of a hot wall at temperature $T_H = 1.0$ featuring hot fins/plates protruding into the enclosure. Different dimensionless times $t^\star$ are reported (see text for details). Panel (b): different geometrical configurations used in the simulation. Panels (c), (d): closeup on the geometrical structure of the fins/plates used.

of sound [28,29]. The thermal dynamics can be easily incorporated into an LBM solver by coupling to f_i a *second*, identical LBM equation where $g_i \leftrightarrow f_i$, $T \leftrightarrow \rho$, $\kappa \leftrightarrow \nu$ and S_i encodes the melting [5,21,22]. Such LBM-based solver is capable of simulating the convective flows coupled to melting dynamics in the presence of complex boundary conditions [5,21,22]. Complex boundary conditions can easily be implemented by imposing them on the populations [21,28,29], by either fixing them at the equilibrium (for sources) or using the halfway bounce-back rule (for no-slip and adiabatic walls) [29]. A site is considered solid, and thus undergoes only thermal dynamics (g_i), until $\phi < 0.5$, at which point the evolution of f_i starts.

Geometry and setup. – Different geometrical configurations of the PCM cell are analyzed. The layouts are systematically selected in order to isolate the role of geometry while keeping constant the total heated area and source temperature T_H, aiming to provide a consistent three-dimensional framework to systematically assess widely used designs in the literature [18,34–36]. An initial group of configurations stems from the simple idea of repeating the same fin a given number of times: we chose four, as it is not too high (hence leading to too complex layouts) nor too low, allowing the agile design of non-trivial three-dimensional layouts. Different geometrical configurations are then obtained via translation: i) *Line*, with the 4 fins horizontally aligned at the same

height; ii) *Rect*, with two fins at the same height, the other two higher at the same horizontal position; iii) *Stagger*, similar to Rect, but with a slight horizontal offset for the higher fins; iv) *Star*, with two fins at the same height, the other two at the same horizontal position, halfway between the other two, one higher and one lower. To better understand the role of the relative closeness of the fins, we devised v) *Square*, a configuration in which the fins are at a larger, equal distance between themselves. The length, height and width (see fig. 1(c)) for a single fin are $(l, h, w) = (63, 4, 8)/L$.

Heat transfer in three-dimensional multi-fin layouts can be influenced by many aspects, such as flow obstruction, conductive path length, orientation relative to gravity and distribution of the heat exchange surface. In this letter we chose to focus on the latter, by comparing the finned configurations with a configurations presenting a single plate, a slab-like ledge, with a total heating surface equal to that of the fins. Given the heating surface of the fin, $S_{\mathrm{H,FIN}} = 2(lw + lh + wh) - wh$, the heating surface of the plate is taken as $S_{\mathrm{H,PLATE}} = 4 \times S_{\mathrm{H,FIN}}$. In these computations, we have excluded the contact surface with the eclosure's wall. The dimensions of the plate (see fig. 1(d)) were then chosen as $(a, b, c) = (56, 5, 48)/L$. The inclusion of the plate was studied in two configurations: vi) *Midplate*, with the plate positioned at about half the height of the enclosure and vii) *Lowplate*: similar to *Midplate*, but at a lower vertical position, in order to test the importance of the latter [15] in 3D environments. Finally, to systematically assess the impact of the presence of the fins/plate compared to a finless wall we have considered two further layouts: viii) *Finless*, a simple cubic enclosure, with no fins, and only the side wall of area L^2 as the source; ix) *Finlesshotter*, similar to *Finless*, but with a source temperature T_{HFL}, chosen in such a way that the total heat exchange is equivalent to a configuration with fins, *i.e.*, $T_{\mathrm{HFL}} \times L^2 = T_{\mathrm{H}} \times (L^2 + 4 \times S_{\mathrm{H,FIN}})$.

The performance of the different geometrical configurations is analyzed by also changing the relevant non-dimensional numbers governing the phenomena under investigation. To tune the intensity of buoyancy forces with respect to diffusive/dissipative effects, the input parameter is the Rayleigh number Ra,

$$\mathrm{Ra} = \frac{g\alpha L^3}{\nu\kappa}(T_{\mathrm{H}} - T_{\mathrm{C}}), \tag{6}$$

while the Stefan number

$$\mathrm{St} = \frac{C_p}{L_f}(T_{\mathrm{H}} - T_{\mathrm{C}}) \tag{7}$$

accounts for the ratio of sensible to latent heat. The Prandtl number is fixed at $\mathrm{Pr} = \nu/\kappa = 1.0$. To monitor the melting process within the cell we take the space integral of the liquid fraction normalized to the PCM volume $V_{\mathrm{PCM}} = L^3 - 4 \times whl$,

$$\langle\phi\rangle(t) = \frac{\sum_{\vec{x}} \phi(\vec{x}, t)}{V_{\mathrm{PCM}}}. \tag{8}$$

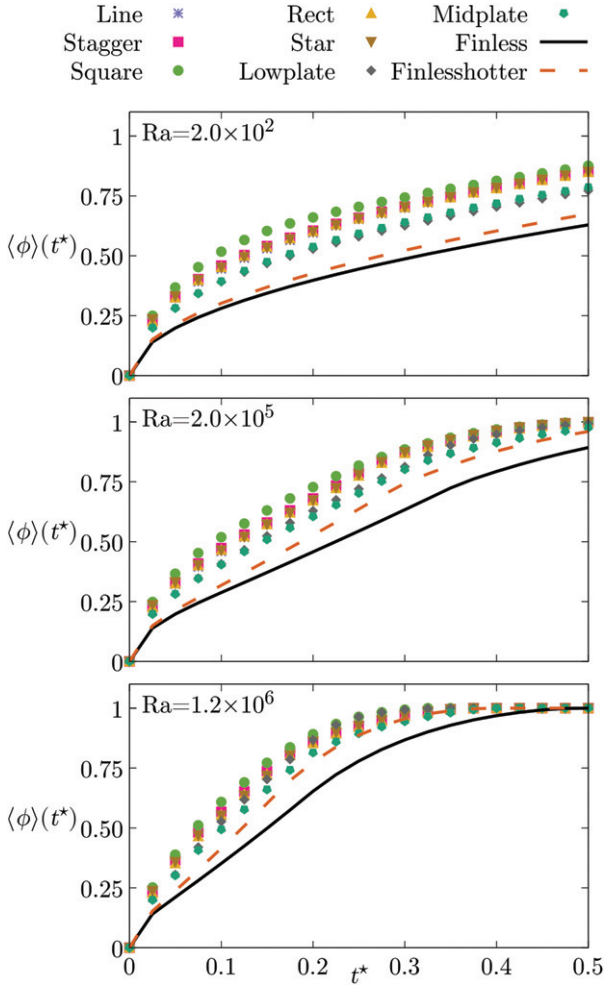


Fig. 2: We report the normalized liquid fraction (cf. eq. (8)) as a function of the dimensionless time $t^\star$ for different Rayleigh numbers (cf. eq. (6)) for a fixed Stefan number $\mathrm{St} = 10.0$. The different geometrical configurations are illustrated in fig. 1 with the exception of Finlesshotter whose details are given in the text.

Finally, for the sake of simplicity, the results are reported using the dimensionless time $t^\star = t/t_{\max}$, with $t_{\max} = 2.0 \times 10^4$ for all the simulations. The other parameters are fixed, in lattice units, at $g = 9.81$, $\nu = 0.1$, $\kappa = 0.1$, $L = 128$, $T_{\mathrm{H}} = 1.0$, $T_{\mathrm{C}} = 0.0$, $T_0 = 0.5$, $T_{\mathrm{HFL}} = 1.39$, $L_f = 1.0 \times 10^3$. In this study, we first focus on three convective cases: $\mathrm{Ra} = 2.0 \times 10^2$, 2.0×10^5, 1.2×10^6 ($\alpha = 1.0 \times 10^{-7}, 1.0 \times 10^{-4}, 6.0 \times 10^{-4}$) . The Stefan number is fixed at $\mathrm{St} = 10.0$ ($C_p = 1.0 \times 10^4$). Then, we focus on a conductive regime ($\mathrm{Ra} = 0.0$, *i.e.*, $\alpha = 0.0$) for two different values of the Stefan number $\mathrm{St} = 1.0, 10.0$ ($C_p = 1.0 \times 10^3, 1.0 \times 10^4$).

To assess the early difference in performance between the fins and the plate we can compute the difference in molten substance $\Delta V_{\mathrm{m}} = 4V_{\mathrm{m,fin}} - V_{\mathrm{m,plate}}$. Choosing $t = 0$ for the sake of simplicity, assuming that the substance melts at a characteristic distance d from the heating

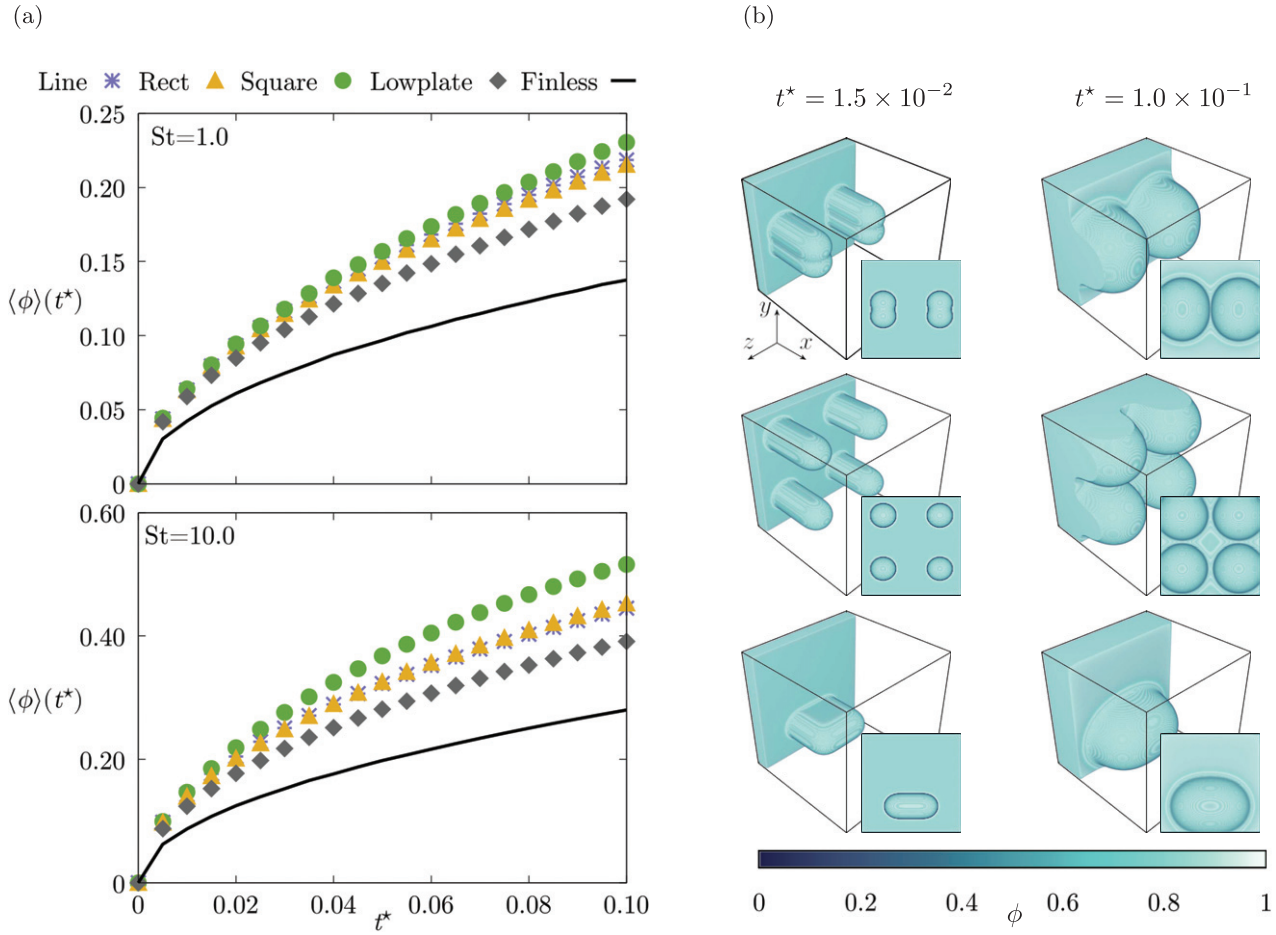


Fig. 3: Analysis of the early-stage melting dynamics in the conductive case, corresponding to a zero Rayleigh number (cf. eq. (6)). Panel (a): we report the normalized liquid fraction (cf. eq. (8)) as a function of the dimensionless time $t^\star$ for different geometrical configurations and different Stefan numbers (cf. eq. (7)). The geometrical configurations used are illustrated in fig. 1. Panel (b): 3D snapshots of the liquid fraction $\phi(\vec{x})$ at selected times from panel (a) for the *Rect* and *Square* configurations, St = 10.0. Each plot includes a front facing perspective on the bottom right.

surfaces and neglecting the melting from the side boundary (reasonably equal in both cases) we get the volumes

$$V_{\mathrm{m,fin}} = (d+l)(2d+h)(2d+w) - lhw, \tag{9}$$

$$V_{\mathrm{m,plate}} = (d+a)(2d+b)(2d+c) - abc, \tag{10}$$

where lhw and abc are the volumes of the heating elements (we are excluding the wall-facing surface). Substituting L and the chosen fin and plate dimensions we get

$$\Delta V_{\mathrm{m}} = \frac{3}{64} d^2 (256d + 129). \tag{11}$$

Results and discussions. – First, we focus on the characterization of $\langle\phi\rangle(t^\star)$ (eq. (8)) as a function of Ra. Figure 2 reports the trends of $\langle\phi\rangle(t^\star)$ for the different geometrical configurations illustrated in fig. 1(b) for $\mathrm{Ra} = 2.0 \times 10^2,\ 2.0 \times 10^5,\ 1.2 \times 10^6$, St = 10.0. Increasing Ra results in better performance for all the configurations in terms of reduced time to complete the melting process ($\langle\phi\rangle = 1.0$). This observation is in line with the literature, as it is known that higher values of Ra increase the importance of convection, which is the main driver of heat transfer in melting systems [37–39]. Moreover, simply raising the source temperature proves less effective than inserting fins, as *Finlesshotter* is generally worse-performing than the finned configurations, proving the fundamental role of the geometry. Increasing Ra can reduce the advantage of the finned configurations with respect to other configurations, as the fins can impede convection by breaking up macroscopical flows; in fact, a *proper* positioning of the fins in the lower parts of the enclosure can favor melting by promoting the onset and development of convection [15]. This can be observed in the plated configurations, as we can see that *Lowplate* performs generally better than *Midplate*, especially at higher Ra, thanks to the development of larger convective structures due to the favorable lower positioning [15]. The performances of plated and finned configurations in fig. 2 are evidently different: we argue that the finned configurations use the *same* heating surface more efficiently, melting more substance in the same time. This is also what was expected from the positivity of ΔV_m in eq. (11). We must stress that the derivation

leading to eq. (11) is valid only in the early stages of melting, where conduction dominates the dynamics. Nevertheless, the fact that the finned configurations produce a larger molten region in the early stages increases the effective liquid-solid interface. This, in turn, promotes heat transfer and leads to a sustained advantage in the overall melting dynamics. We note that by increasing Ra, the plated configurations get closer to the finned ones, *i.e.*, the former gain more (in terms of melted PCM) compared to the latter; arguably, this is due to the presence of a larger surface $\perp \vec{g}$, since we have $ac > 4lw$ for the chosen sizes.

Another important feature observable in fig. 2 is that configurations i)–iv), despite being quite different geometrically, deliver essentially the same performance. Configuration v) (Square) with further apart fins, performs better than i)–iv): this suggests the presence of effects triggered by the closeness of the fins that may lead to an energetic overlap in molten zones, during the early stages, thus "wasting" thermal energy on already melted substance. In order to provide some quantitative evidence of this hypothesis, we focus on a purely conductive regime ($\mathrm{Ra} = 0$). Moreover, since the melting speed in the conductive dynamics is regulated by the Stefan number, we analyze data related to two different values, $\mathrm{St} = 1.0, 10.0$, aiming to prove the robustness of the trend. In fig. 3(a) we analyze the conductive dynamics of $\langle\phi\rangle(t)$ for different geometrical configurations. A lower value of St delivers slower melting, but the separation happens regardless, just delayed in time. Again, the plated configurations perform worse. Figure 3(b) expands on the preceding panel by showing snapshots of $\phi(\vec{x})$ for $\mathrm{St} = 10.0$ for *Rect* and *Square* at two relevant times from fig. 3: the time for which *Square* visibly separates from the other configurations ($t_1^\star = 1.5 \times 10^{-2}$) and at the last time shown ($t_2^\star = 1.0 \times 10^{-1}$). A comparison at $t_1^\star$ between the first and second rows highlights the presence of the interference in molten zones in the *Rect* configuration, and its absence in *Square*. This overlap is clearly responsible for the reduction of the total molten volume in the *Rect* configuration with $\langle\phi\rangle(t_1^\star) = 1.73 \times 10^{-1}$, $\langle\phi\rangle(t_2^\star) = 4.53 \times 10^{-1}$ *vs.* $\langle\phi\rangle(t_1^\star) = 1.86 \times 10^{-1}$, $\langle\phi\rangle(t_2^\star) = 5.16 \times 10^{-1}$ of *Square*. In the third row, *Lowplate* is shown, giving visual proof of its worse spatial efficiency ($\langle\phi\rangle(t_1^\star) = 1.73 \times 10^{-1}$, $\langle\phi\rangle(t_2^\star) = 3.91 \times 10^{-1}$). Overall, the better performance of *Square* proves that the relative distance between fins matters: further apart fins gain an advantage that cascades in overall better melting.

Conclusions. – Phase change materials (PCMs) are playing an increasingly important role in energy-related applications, ranging from thermal energy storage and building technologies to thermal management in electronics and hydrogen systems [5,40–42]. Improving PCM heat transfer capabilities without introducing active components is a key technological challenge, making passive optimization strategies particularly attractive for engineering applications. The insertion of fins inside a PCM is known to be a simple and effective way of enhancing the overall thermal performance [34]. This letter systematically addressed the impact of fin insertion on melting in a selection of geometrical layouts and hydrodynamical configurations. First, two aspects already known from the literature [15,34] were confirmed: 1) fin insertion provides more efficient melting with lower relative temperature due to the onset of convection and 2) fins positioned in the lower parts of the PCM give rise to larger convective motions (and, thus, enhanced thermal transfer) than equivalent fins positioned in the upper part of the domain. We further observed that distributing the same heating surface over multiple slender fins is more effective than concentrating it into a single plate-like element, as multiple sources melt more PCM in the early stages, leading to better performance. From the analysis of the fin layouts, we found robust evidence that the excessive closeness of fins can lead to overlapping molten zones, resulting in inefficient use of thermal energy. In contrast, configurations with more distant fins are characterized by a smoother distribution of the heat, leading to efficient melting. This analysis was specifically possible thanks to the extension to three dimensions, which allowed us to study the effect of the interstitial spaces on the plane $\perp \vec{g}$ and more generally granted access to more "degrees of freedom" in positioning and design of fins. The findings in this letter hint at interesting general trends related to the geometry of multi-fin arrays. If from one side the argument invoked on the overlapping of molten zones appears as quite generic and robust, the present preliminary study was limited to a selection of configurations; hence, an interesting and natural next step would be conducting investigations in order to assess whether the observed trends remain valid in a wider range of values of Ra, St and other physical parameters. Furthermore, testing different sizes/shapes for the designed fins, as well as studying more in detail the role of interstitial spaces in relation to the fin height, could reveal more interesting peculiarities on melting dynamics. Finally, the present analysis could be extended towards broader design principles by exploring more complex fin geometries and introducing hybrid technical solutions (metal foams, nanoparticle insertion and different chemical compositions among the others) to improve thermal transfer [11,12,19,43,44].

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Data availability statement: The data that support the findings of this study are available upon reasonable request from the authors.

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